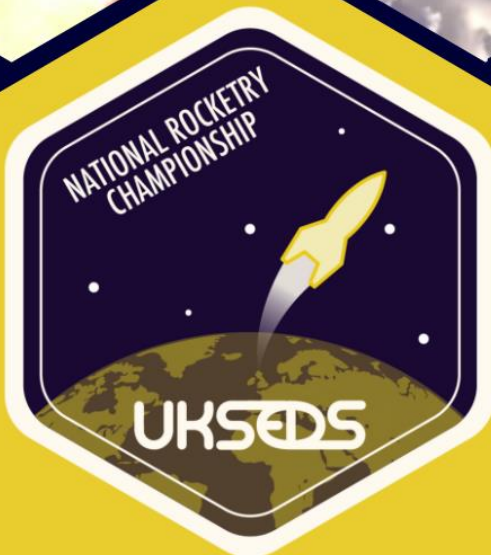




NATIONAL ROCKETRY CHAMPIONSHIP 2025-26

Critical Design Review

<INSERT TEAM NAME>



SUPPORTED BY



UNITED KINGDOM
ROCKETRY
ASSOCIATION



UKSEDS National Rocketry Championship Critical Design Review

2025 – 2026

UKC AstraNova



Issued by (Project Lead): Cara Hefer

University: University of Kent

Table of Contents

1. Introduction	5
1.1. Mission Statement	5





1.2. Objectives	6
2. Project Management	6
2.1 Team Members	6
2.2 Financial Budget	7
2.3 Project Risks and Mitigations	7
2.4 Project Plan	9
2.5 Work Breakdown Structure	9
3. Preliminary Concept	9
3.1. Launch Vehicle	9
3.1.1. Launch Vehicle Overview	10
3.1.2. Concept Of Operations (CONOPS)	11
3.1.3. OpenRocket Simulation	11
3.1.4. Recovery System Design	15
3.1.5. Launch Vehicle Components	17
3.2. Rideshare Payload	18
3.2.1. General Overview	19
3.2.2. PCB Schematics	21
3.2.3. System Integration	22
4. Systems Engineering	23
4.1 Requirement Compliance Analysis	23
4.2 Budgets	27
5. Manufacturing Processes	27
5.1 Launch Vehicle	27
5.2 Rideshare Payload	28
5.3 Issues and Solutions	28
6. Testing	28
6.1 Test Matrix	28
6.2 Test Results	29
7. Final Design	29
7.1 Design Changes	29





7.2 Improvements and Future Work	30
8. Acknowledgements	31
9. References	31

1. Introduction

UKC AstraNova is a rocketry team from the University of Kent. We're building a rocket for the UKSEDS National Rocketry Championship with the goal of reaching 670 m, carrying both a customer payload and our own payload, and landing safely within 1 km of the launch site.

The team is made up of students covering propulsion, structures, avionics, recovery, payload, and systems:

1. Cara-Alexandra Hefer – Team Lead / Systems
2. Brittany Nsongo – Vice Team Lead / Structures
3. Alexander Barry Dee – Propulsion
4. Arda Aydinlar – Systems
5. Theodoros Zilios – Recovery
6. Adam McGrath – Recovery
7. Calum Lewis – Avionics
8. Jack Oliver Shorey – Propulsion
9. Lucian Willcock – Payload
10. Robert Charles Langley Parry – Structures

The rocket is designed to reach the target altitude, deploy both payloads, and come back down safely using a controlled recovery system.

1.1. Mission Statement

To design, build, and fly a rocket that reaches 670 m, delivers its payloads successfully, and lands safely within the recovery zone.

1.2. Objectives

Launch the rocket to a target altitude of 670 m.

Successfully deploy the recovery system.

Recover the rocket safely within 1 km of the launch site.





Recover both the customer payload and team payload intact.
Successfully record and recover flight data from the avionics system.
Verify stable flight and controlled descent throughout the mission.
Ensure all systems pass ground testing before launch.
Recover the rocket in a reusable condition where possible.
Complete the flight in full compliance with UKSEDS NRC safety rules.

2. Project Management

2.1 Team Members

Name	Role	Year/Course
Cara-Alexandra Hefer	Team Lead/Systems Engineer	1 st Year/Astronomy, Space Science and Astrophysics
Brittany Nsongo	Vice Team Lead/Structural Engineer	1 st Year/Astronomy, Space Science and Astrophysics
Alexander Barry Dee	Propulsion Engineer	2 nd Year/Physics with Astrophysics
Arda Aydinlar	Systems Engineer	2 nd Year/Physics with Astrophysics
Theodoros Zilios	Recovery Technician	1 st Year/Mechanical Engineering
Adam McGrath	Recovery Technician	2 nd Year/Physics
Calum Lewis	Avionics Engineer	2 nd Year/Electrical and Electronic Engineering
Jack Oliver Shorey	Propulsion Engineer	4 th Year/Physics with Astrophysics
Lucian Willcock	Payload Specialist	1 st Year/Physics with Astrophysics
Robert Charles Langley Parry	Structural Engineer	2 nd Year/Ecology and Conservation

Figure 1: Table of Team Members





Figure 2: Image of Team Members

2.2 Financial Budget

We have been allocated a budget of £500 for this competition. Our current and future expenses are in a neat table format as follows:

Name	Amount
2 x BMFA membership	£102.00
Motor	£30.00
Shock cords	£25.00
Parachute	£35.00
PCB Design	£60.00
Extra Funds (in case)	£100.00
Total	£352.00

Figure 3: Table of Budget

2.3 Project Risks and Mitigations

Identify the major risks associated with your project and rocket design, and how you expect to manage these in the build-up to Launch Day.

Risk	Impact	Mitigation
Motor failure or underperformance	Rocket fails to reach target	Use manufacturer-certified motors only, follow motor storage and handling



	altitude or becomes unstable	guidelines, and select motor based on simulation and safety margins
Recovery system failure	Rocket impacts at high speed and is damaged or lost	Ground-test ejection charges, use redundant deployment methods where possible, pack parachutes carefully, and perform full recovery system checks before launch
Avionics failure (no data or no deployment trigger)	Loss of flight data and/or recovery system not triggered	Bench-test avionics, perform vibration and continuity checks, carry spare components, and use simple, proven deployment logic
Structural failure during ascent	Rocket breaks apart in flight	Use safety factors in structural design, test joints and airframe sections on the ground, and inspect for damage after every test
Stability issues (poor CG/CP margin)	Rocket becomes unstable or unsafe in flight	Simulate flight in OpenRocket, verify centre of gravity with motor and payload installed, and perform swing or rail tests
Payload damage or detachment	Loss of customer payload or data	Secure payload with mechanical restraints, test payload bay under vibration, and include padding or shock isolation
Landing outside recovery zone	Rocket lost or hard to retrieve	Tune descent rate, choose appropriate parachute size, track descent using beeper/GPS if allowed, and account for wind conditions
Human error during assembly	Miswiring or incorrect installation	Use checklists, peer-check critical steps, and perform final pre-launch inspections
Limited testing opportunities	Design issues not caught before competition	Do extensive ground testing on campus, subsystem tests (ejection, avionics, fit checks), and dry runs of full assembly



On-campus build/testing constraints	Incomplete testing of full system	Focus on safe subsystem testing indoors (avionics, recovery packing, structural fits) and document all procedures clearly
Schedule slip / team availability	Rushed build or missed deadlines	Set internal deadlines, assign clear subsystem owners, and hold regular progress checks
Weather on launch day	Unsafe launch or drift out of recovery zone	Prepare multiple parachute sizes if allowed, plan for wind limits, and be ready to scrub or adjust configuration

Figure 4: Table of Risk Assessment

2.4 Project Plan

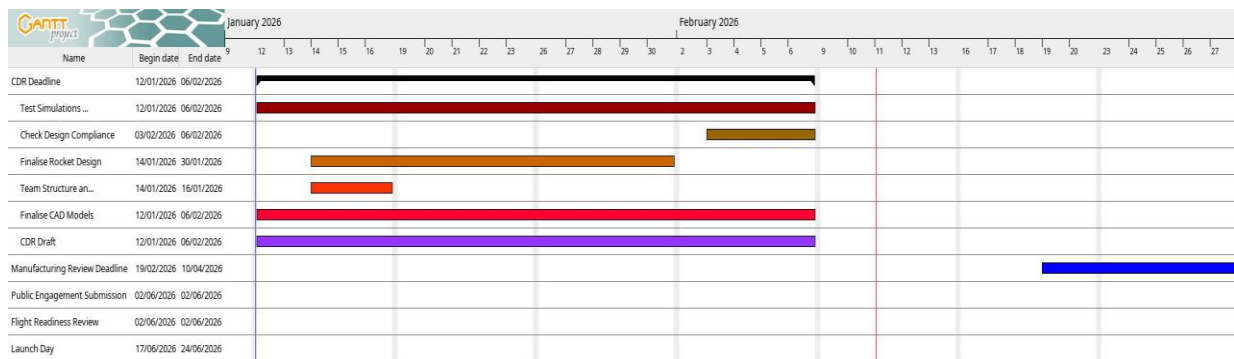


Figure 5: Image of Gantt Chart

2.5 Work Breakdown Structure

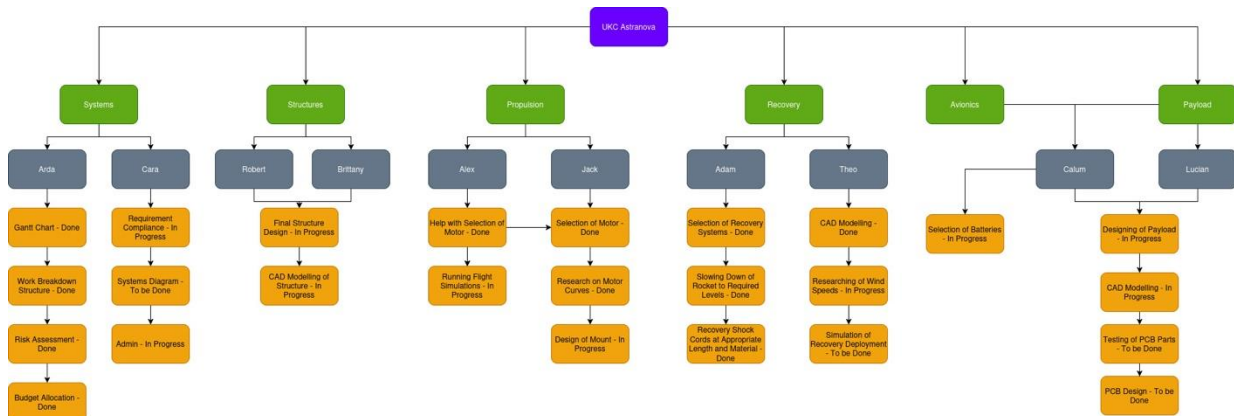


Figure 6: Image of Work Breakdown Structure

3. Preliminary Concept

3.1. Launch Vehicle

3.1.1. Launch Vehicle Overview

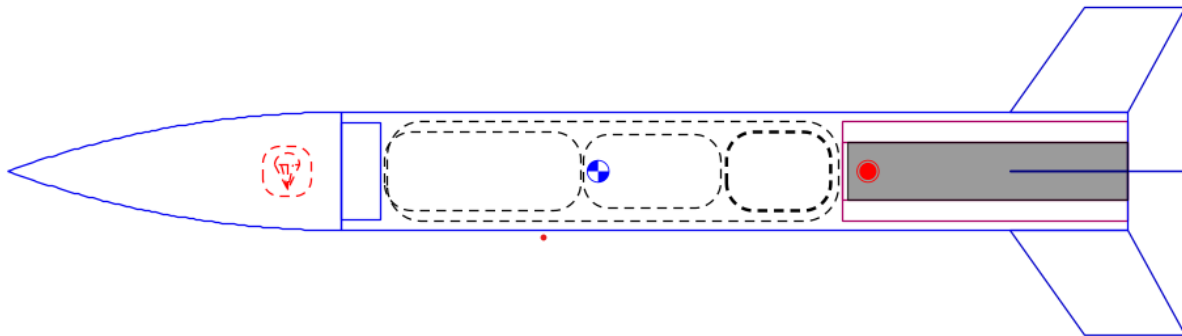


Figure 7: Diagram of Rocket Design

The rocket uses a 4-fin design which are fixed at the bottom of the launch vehicle (NRC RS-TSN-SAR-001). They use a trapezoidal shape. They are made of spruce wood and have an aerofoil cross-section (NRC RS-TSN-SAR-002 & NRC RS-TSN-SAR-003). The motor (NRC RS-MRD-MIS-004) is secured by a 3D-printed tube which will hold the motor in place and will be secured using 5mm screws to the main body (NRC RS-TSN-MOR-001). The tube will also act as a bulkhead between the motor and the avionics compartment.

The payloads are secured in one large 3D printed part, containing the customer's payload at the bottom, the rideshare payload (NRC RS-MRD-MIS-007) at the top, and batteries stored in the middle.

The main body is made using balsa wood veneers fixed using wood adhesives (NRC RS-TSN-SAR-004). The internal width of 5cm and a thickness of 0.5cm. It has a length of 40cm.

The nosecone will be designed using 3D CAD and printed using lightweight plastic filament. The nose cone uses an Ogive shape. It will be held to the main body using a tight-fitting shoulder design.

The recovery system will be held separate from the avionics and stored partly in the nose cone (NRC RS-TSN-RRS-001).

The total length of the rocket is 59.9cm, and a total weight of 955g.

3.1.2. Concept Of Operations (CONOPS)

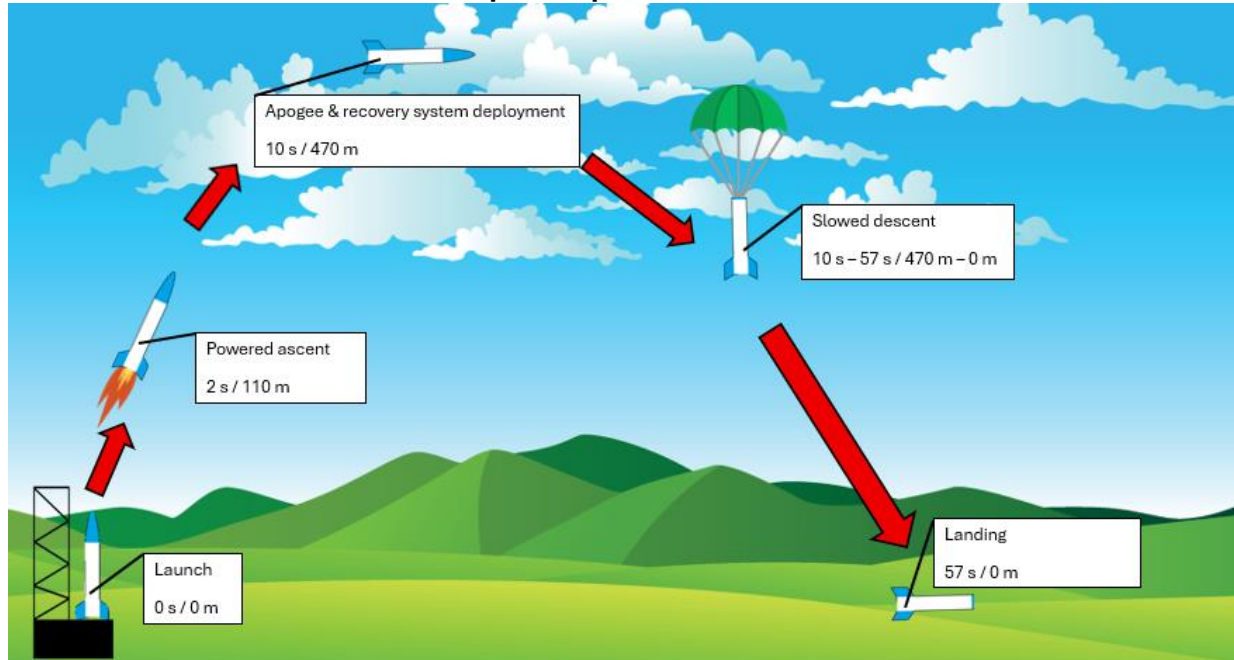


Figure 8: Diagram of Order of Operations.

3.1.3. OpenRocket Simulation

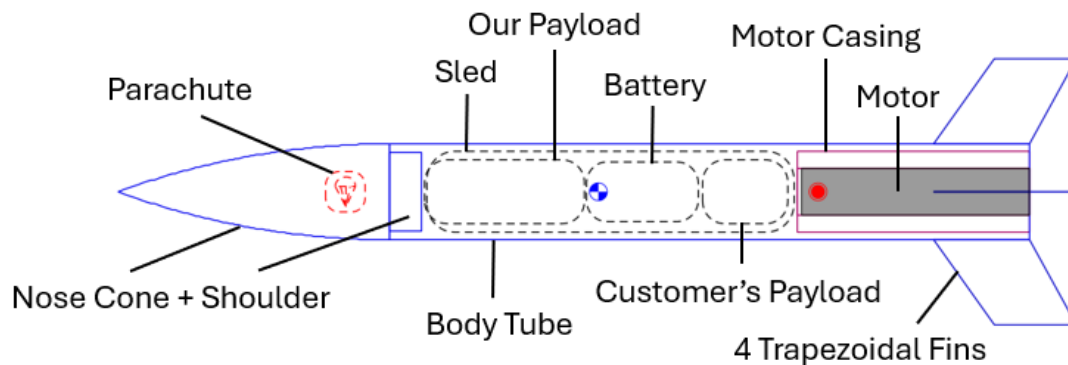


Figure 9: Annotated diagram of our design in OpenRocket.

Flight simulations were carried out using OpenRocket 24.12, as required by UKSEDS, to estimate the vehicle's ascent, stability throughout the flight, and recovery behaviour. The model included the final rocket structure provided by the structures team, measured mass properties, and the manufacturer-provided thrust curve for the selected motor, the Pro29 108G68-13A from Cesaroni Technology. Simulations were run using a 2 metre launch rod to match the expected launch setup. A windspeed of 0 m/s was used to isolate propulsion-driven vehicle behaviour, and additional wind cases, 10 m/s and 15 m/s, were analyzed to assess drift and recovery constraints. The

results were used to verify that the vehicle meets altitude, stability, and landing requirements.

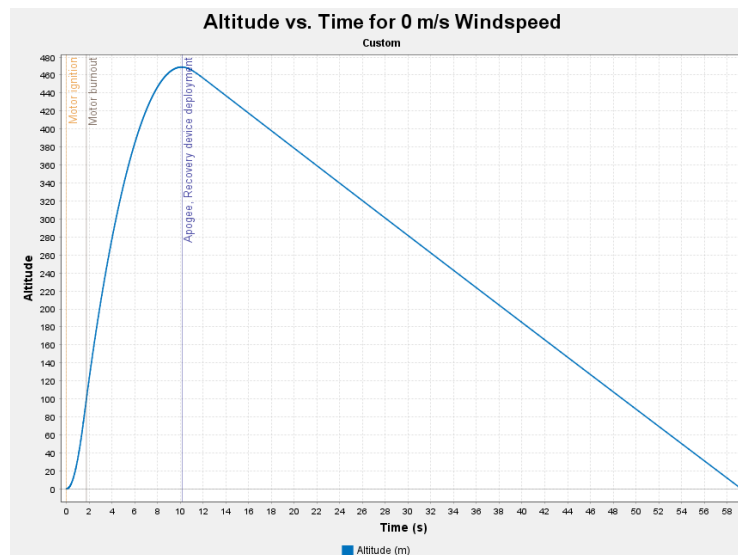


Figure 10: Graph showing altitude against time for the rocket's ascent and descent with no wind effects.

Figure 10 shows the simulated altitude versus time graph under 0 m/s wind conditions. The vehicle experiences a rapid increase in altitude during the motor burn phase, reaching burnout at approximately 2 seconds. This matches the manufacturer-provided thrust curve given. Following burnout, the rocket enters a coast phase where vertical velocity gradually decreases due to gravity and drag. Apogee occurs at approximately 10 seconds at a maximum altitude of roughly 470 metres. This equates to ~1540 ft, while this does not reach the maximum 2200 ft, it obtains many of the marks awarded for altitude while not going over the required distance. This satisfies the requirements NRC RS-MRD-MIS-001, NRC RS-MRD-MIS-002, and NRC RS-MRD-MIS-003. Recovery deployment is triggered at apogee, resulting in a smooth transition to descent. The nearly linear descent profile indicates a stable parachute deployment and a consistent descent rate until landing at approximately 58 seconds. The predicted apogee remains within the required altitude constraints and demonstrates that the selected propulsion system achieves the intended performance expected.

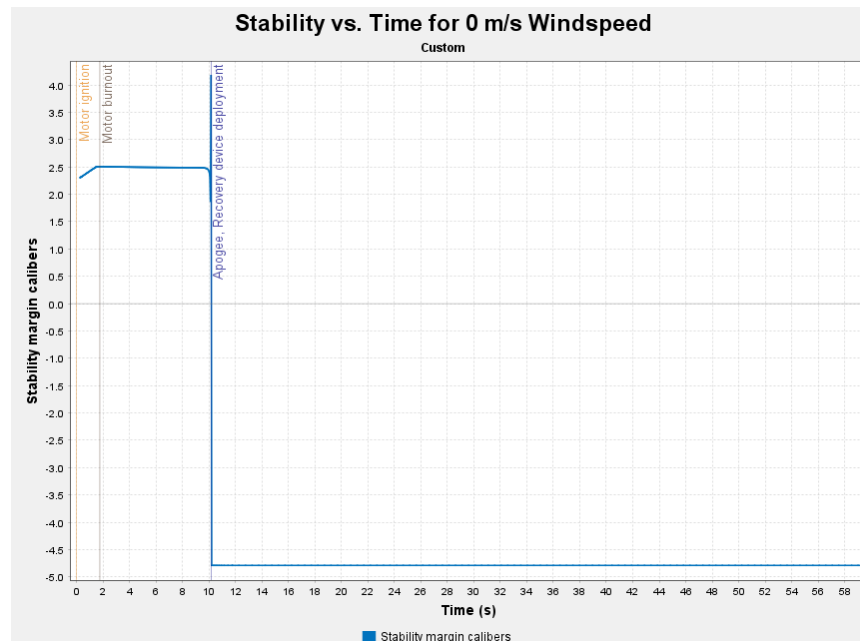


Figure 11: Graph showing stability against time for the rocket's ascent and descent with no wind effects.

Figure 11 shows the simulated stability margin versus time graph under 0 m/s wind conditions. At ignition, the vehicle has a stability margin of approximately 2.3 calibres, increasing slightly during motor burn as propellant mass decreases. Throughout powered ascent and coast, the stability margin remains within 2.3-2.5 calibres, satisfying the minimum stability requirement, NRC RS-TSN-SMN-003, and indicating a consistently stable design. No significant reductions in stability are observed during the burn phase, demonstrating that the centre of gravity shift does not compromise vehicle stability. Following recovery deployment at apogee, the stability margin becomes negative as the vehicle transitions to parachute descent; this behaviour is expected and does not impact ascent stability performance.

For the drift constraints we simulated the data using wind speeds of 10 m/s and 15 m/s and compared the historic values of windspeed in the area which we got through visualcrossing.com [1], these values are the average wind speeds that have occurred on the two launch dates (5.4m/s on the 11th and 4 m/s on the 18th of June) from the past 10 years. this was to get an idea for the likelihood of each outcome. wind direction has been set to 180 deg from 90 deg for better graph clarity.

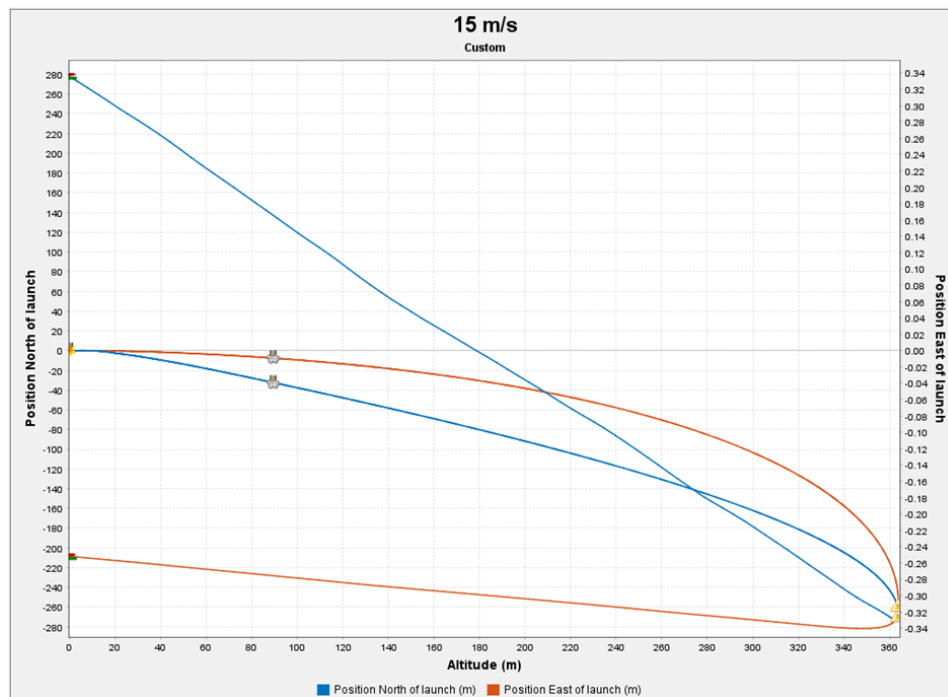


Figure 12: Graph showing displacement from launch with 15 m/s wind speeds.

Figure 12 shows the graph of the rocket's longitudinal and lateral displacement against altitude when the windspeed is at 15m/s. In this case the rocket lands 278.1 meters north and 0.25m west from the launch position which is well within the bounds of the requirements. However, the rocket lands with a speed of 17.4 m/s which is above the required landing speed however we chose to overrule this since windspeeds of 15m/s is an extreme and unlikely case.

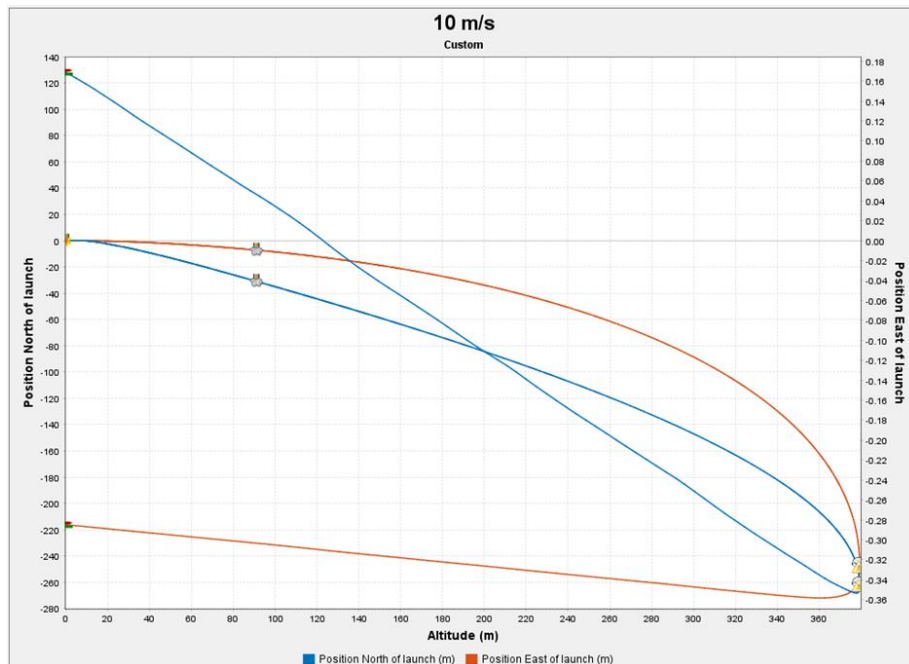


Figure 13: Graph showing displacement from launch with 10 m/s wind speeds.

Figure 13 shows the graph the graph of the rocket's longitudinal and lateral displacement against altitude when the windspeed is at 10m/s. here the rocket lands 128.26m north and 0.28 m west from the landing position which is also within the bounds of the requirements; but this time the rocket lands of 13.3 m/s which is in the bounds of the requirements

In both cases the parachute ejects at a speed of 27.7 m/s due to the extreme wind changing the trajectory of the rocket to be flatter and causing the parachute, which is set to eject at apogee, to eject when the rocket is moving horizontally. However, both cases seem very unlikely.

3.1.4. Recovery System Design

The type of recovery system:

The recovery system to be used will be the TAC-24 parachute from Giant Leap (NRC RS-TSN-RRS-001). This parachute has been tested in OpenRocket to ensure that it will slow the rocket down to the required amount, >15m/s, and this is shown in our weight / descent graph (NRC RS-TSN-RRS-002 , NRC RS-TSN-RRS-003).

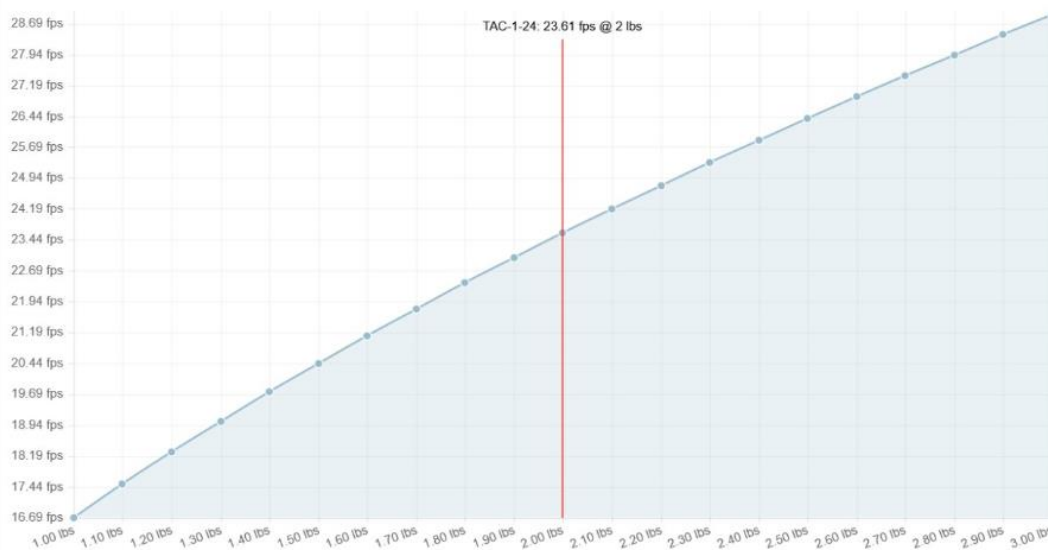


Figure 14: Weight Descent graph

To aid this we will be using a commercial flight computer (NRC RS-TSN-RRS-004) to record and compute the necessary calculations needed to determine when the recovery will be triggered.

Shock cords made of Kevlar and will be used to make sure that no piece of the rocket is detached when the recovery is deployed and all cords will be the appropriate length of over 2m (NRC RS-TSN-RRS-007, NRC RS-TSN-RRS-008).

All electronics related to the recovery system will be isolated from the rest of the rocket (NRC RS-TSN-RRS-005).

Tilt switches will not be used in the design of the recovery system as all calculations will be done by the flight computer (NRC RS-TSN-RRS-006).

How recovery is detected:

Recovery deployment will be controlled by a Mercury VI altimeter and flight computer which will be disabled until the rocket is being prepared for launch (NRC RS-TSN-RRS-007). This device uses barometric pressure sensing and onboard accelerometers to determine key flight parameters, including altitude, vertical velocity, and acceleration. [2]

Apogee will be detected when the flight computer identifies a transition from positive to negative vertical velocity. Upon confirming apogee, the altimeter will trigger the primary ejection charge to initiate recovery deployment. The system will be configured and calibrated prior to launch to ensure accurate pressure readings and reliable apogee detection.



How recovery is triggered:

Recovery will be initiated at apogee and controlled by the flight computer. The onboard flight computer will detect apogee using barometric and/or accelerometer data and trigger deployment of the recovery system.

At apogee, an ejection charge will fire to separate the nose cone from the airframe, allowing the parachute to deploy safely from the forward section of the rocket. The separation event will be controlled by the avionics system to ensure deployment occurs at the correct flight phase (NRC RS-TSN-RRS-005).

Shock cords will connect the nose cone and main airframe to prevent complete separation after deployment. These will be sized appropriately to withstand deployment loads and reduce the risk of structural damage (NRC RS-TSN-RRS-008).

3.1.5. Launch Vehicle Components

The motor selected to deliver thrust for the rocket was the Cesaroni Technology Pro29 108G68-13A. This rocket provides a total impulse of 107.8 newton seconds of thrust with a burn time of 1.61 seconds [3]. This motor falls within the regulation outlined by UKSEDS of having an impulse of less than 130 newton seconds as well as being a 2 grain, 29 millimetre motor [NRC RS-MRD-MIS-004]. It was selected due to its constant in flight thrust, providing a stable and steady burn throughout the duration of the flight without any sudden increases or decreases in thrust as seen by the graph in the figure below [4]. This constant thrust is hypothesised to provide more stability in flight due to the lack of any sudden changes during the main burn phase, as well as providing extra thrust at the beginning of the burn in order to accelerate the rocket to 20 metres per second before it leaves the launch rail. With the current design of the rocket, this motor will be able to provide enough thrust to propel the rocket to an apogee of roughly 470 metres.



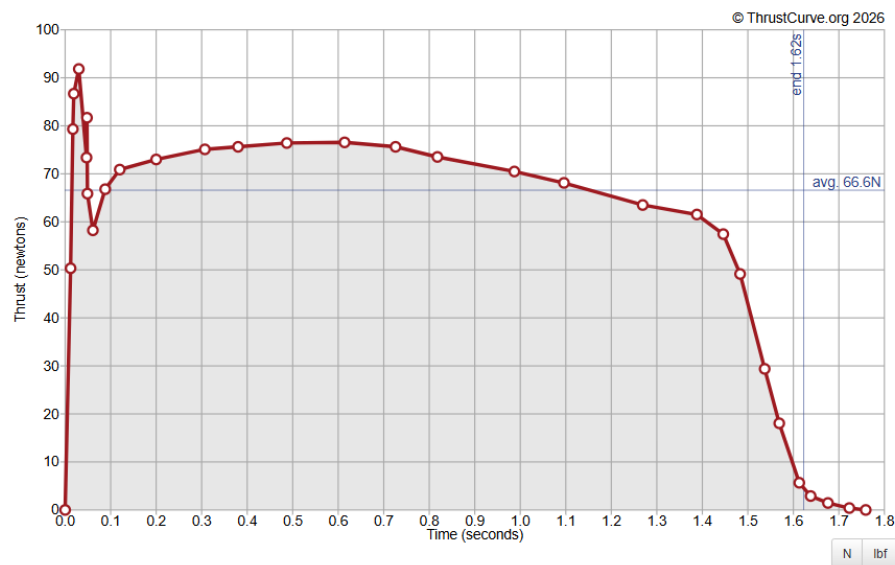


Figure 15: Manufacturer-provided thrust curve.

The motor mount, as seen in the figure below, was designed to accommodate the motor whilst minimising weight. The topmost part of the mount will be screwed into the fuselage (using metal screws) to ensure a secure connection between the motor, mount and the body of the rocket is achieved. The motor will then be placed into the centre of top piece of the mount, and the bottom piece will be threaded on to the top piece using the designed threads. Once threaded on, the bottom will also be screwed to the fuselage. This design will not only effectively secure the motor, but also allow for quick, easy access to the motor and reduce the weight of the mount when compared to other designs.

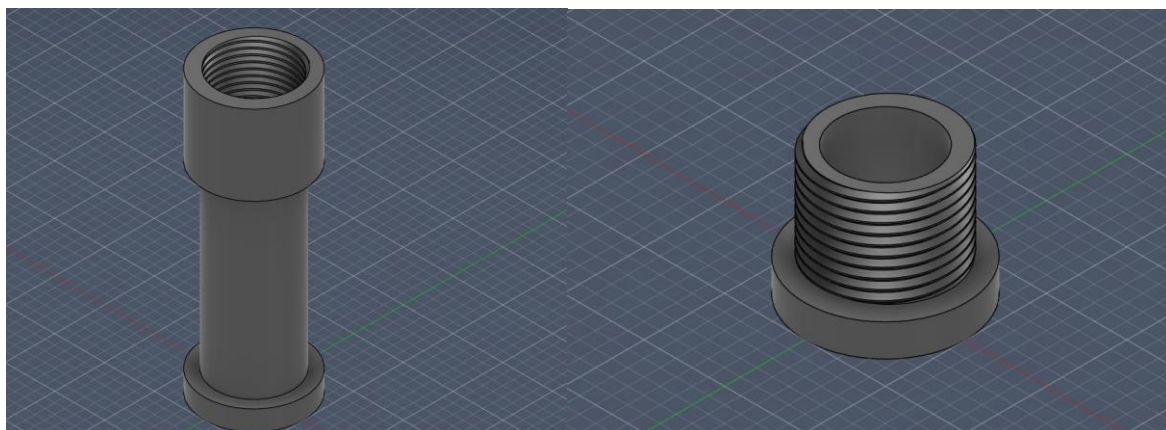


Figure 16: Diagram of Motor Mount

3.2. Rideshare Payload

3.2.1. General Overview

The rideshare payload is a competition-specific avionics system designed to:

- Measure launch vehicle altitude
- Log altitude vs time data
- Display apogee onboard
- Record and log additional independent sensor data

The system consists of:

- Custom-designed PCB (NRC 2025–26 specific)
- Raspberry Pi Pico 2 (SMD mounted)



Figure 17: Raspberry Pi Pico 2

- Waveshare 10DOF IMU (Version D, barometer-based altitude included as potential separate sensor)



• Figure 18: Waveshare IMU

- Waveshare e-paper display

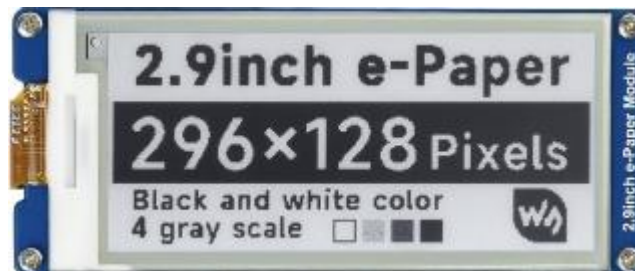


Figure 19: E-paper Display

- microSD storage, Figure 10 was used as a reference

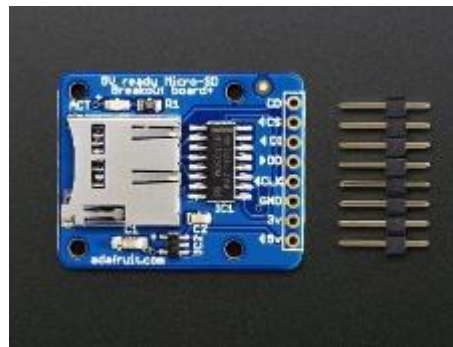


Figure 20: Sd card Reader

- DHT22 temperature sensor



Figure 21: DHT22 Sensor

- Shared 7.4 V 2.25 Ah 2S Li-ion battery
- External arming switch

Altitude is measured using the integrated barometer at **50 Hz**, avoiding inertial drift (RS-MRD-RPD-002).

Apogee is computed onboard and displayed on the e-paper display. E-paper retains its last state without power, ensuring compliance with RS-MRD-RPD-004.

Altitude and timestamp data are written directly to microSD in CSV format. An altitude vs time plot (feet vs seconds) can be produced within 10 minutes of recovery (RS-MRD-RPD-003).

Additional recorded data, which should be written to the MicroSD (independent of altitude sensor) includes acceleration, gyroscope, magnetometer and temperature data, satisfying RS-MRD-RPD-006.

The PCB includes a silkscreened team logo (RS-MRD-RPD-005).

Estimated rideshare mass: ~100 g (50 g PCB + 50 g sled).

3.2.2. PCB Schematics

A dedicated PCB has been designed from scratch for this competition (complying with RS-MRD-RPD-001).

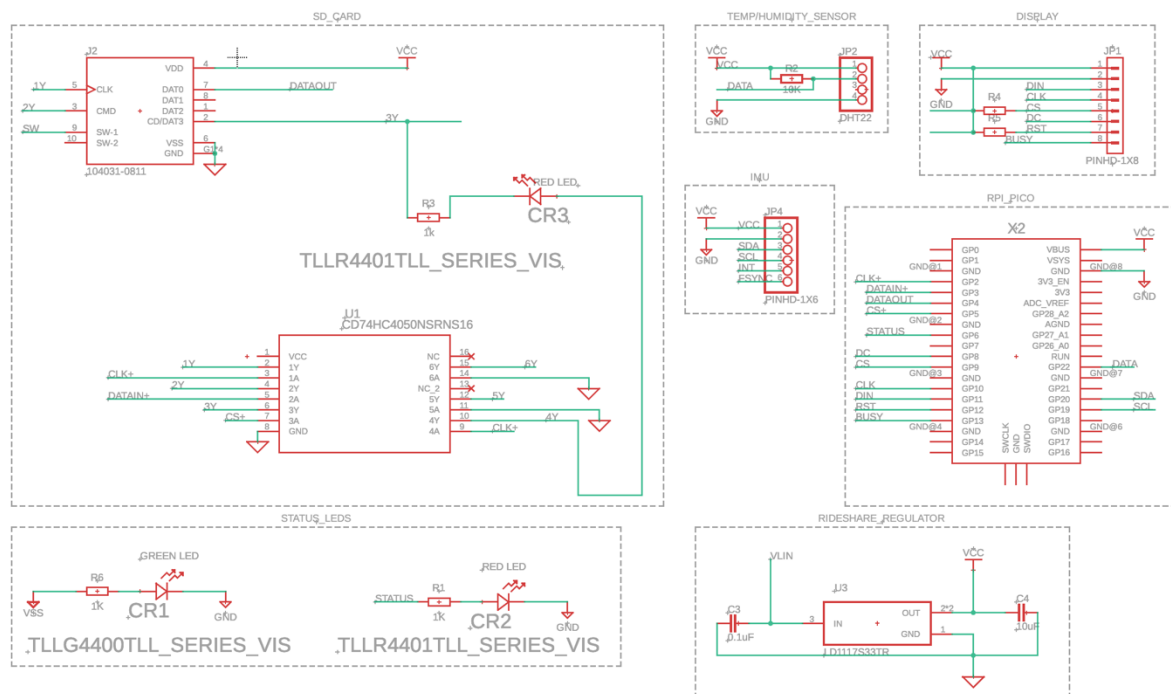


Figure 22: PCB Schematic

Integrated subsystems:

- Raspberry Pi Pico 2 (SMD pads)
- 4 A-rated buck converter (5 V Customer Payload supply)
- LDO regulator for rideshare electronics
- I²C bus for IMU and DHT22
- SPI bus for e-paper and microSD
- Decoupling and required pull-ups

Power architecture:



7.4 V battery → 5 V buck (Customer Payload)

7.4 V battery → LDO → Rideshare electronics

The buck converter exceeds the 1.5 A maximum Customer Payload requirement (RS-MRD-CPD-003), providing margin.

3.2.3. System Integration

Mechanical

Payload bay internal diameter: **50 mm**, exceeding the 40 mm minimum (RS-MRD-CPD-007).

The official UKSEDS Launch Vehicle adapter interface geometry remains unmodified (RS-MRD-CPD-006).

The thrust axis passes centrally through the Customer Payload (RS-MRD-CPD-008).

The payload bay is unpressurised for barometric operation (RS-MRD-CPD-010).

Three M3 bolts are provided for mounting (RS-MRD-CPD-011).

Electronics are mounted on a removable 3D-printed avionics sled to:

- Improve vibration resistance
- Allow rapid access
- Provide cable strain relief

External arming switch for the battery to be added to cut down time needed on the Launchpad to prepare the Rocket. Another thing done to cut down time, is to add an avionics sled with a twist lock mechanism, to quickly be able to take the payloads in and out of the rocket faster, meaning less time on the pad before and after the flight.

Electrical

Customer Payload is powered via a regulated 5 V bus through a female XT30 connector (RS-MRD-CPD-004).

Battery capacity:

$7.4 \text{ V} \times 2.25 \text{ Ah} = 16.65 \text{ Wh}$

Customer Payload requirement (45 min):

$7.5 \text{ W} \times 0.75 \text{ h} = 5.63 \text{ Wh}$





Estimated rideshare consumption: ~1 W

Total required ≈ 6.4 Wh

Energy margin $>2.5\times$ (RS-MRD-CPD-002, RS-TSN-ESS-001).

External arming switch allows system activation without disassembly (RS-TSN-ESS-002).

All wiring is multi-core stranded (RS-TSN-ESS-003).

Recovery is handled by a Mercury V1 flight computer with an independent battery, ensuring electrical isolation.

4. Systems Engineering

4.1 Requirement Compliance Analysis

Requirement ID	Met? [Yes, No, Partially]	Rational
NRC RS-TSN-MOR-001	Yes	The motor mount will transfer thrust being secured to both the body and the motor, allowing for minimal vibrations and prevention of slipping.
NRC RS-TSN-MOR-002	Partially	This is to be tested once we have all the parts
NRC RS-TSN-MOR-003	Partially	This is to be tested once we have all the parts
NRC RS-TSN-SMN-001	No	The rocket achieves a launch speed of 17m/s off the rail
NRC RS-TSN-SMN-003	Yes	The rocket has been designed to maintain a stability margin between 1.87 and 2.5 during all stages of flight
NRC RS-TSN-SAR-001	Partially	The fins are designed to stay in line with the rockets longitudinal axis but has yet to be constructed





NRC RS-TSN-SAR-002	Partially	To be tested once we have the parts
NRC RS-TSN-SAR-003	Partially	To be tested once we have the parts
NRC RS-TSN-SAR-004	Partially	To be tested once we have the parts
NRC RS-TSN-SAR-005	Yes	We are not using interconnected tubes in our design for the main body
NRC RS-TSN-SAR-006	Partially	To be tested once we have all the parts
NRC RS-TSN-SAR-007	Yes	Two rail guides are included in the main body of the launch vehicle
NRC RS-TSN-SAR-008	Yes	Finite Element Analysis was not used to select or test materials
NRC RS-TSN-ESS-001	Yes	The power of the battery to be used will be capable of supporting this.
NRC RS-TSN-ESS-002	Partially	This has not been tested; however, that is the theoretical expectation.
NRC RS-TSN-ESS-003	Yes	A multi-core wire has been included in the design
NRC RS-TSN-ESS-004	No	We're not communicating with the rocket wirelessly
NRC RS-TSN-RRS-001	Yes	Parachute to be used: Spherachute plus+ W/swivel 24". This has been simulated on OpenRocket.
NRC RS-TSN-RRS-002	Yes	OpenRocket simulation shows the system shall reduce the speed to 9.5m/s
NRC RS-TSN-RRS-003	Partially	Research is being done to estimate this





NRC RS-TSN-RRS-004	Yes	The Mercury V1 is UKRA approved.
NRC RS-TSN-RRS-005	Yes	A bulkhead is in between recovery system and electrical system
NRC RS-TSN-RRS-006	Yes	Flight computer is being used
NRC RS-TSN-RRS-007	Partially	Uncertainty of this for the moment. Awaiting part to confirm.
NRC RS-TSN-RRS-008	Yes	Holes to be created in the nozzle to secure the eye bolts using the nut.
NRC RS-TSN-RRS-007	Yes	We have a role of shock cords to customize the size
NRC RS-TSN-RRS-008	Yes	They're kevlar.
NRC RS-MRD-MIS-001	No	The OpenRocket simulation reached an apogee of 470m.
NRC RS-MRD-MIS-002	No	We don't reach beyond the target height.
NRC RS-MRD-MIS-003	Partially	To be tested
NRC RS-MRD-MIS-004	Yes	Motor fits in the guidelines
NRC RS-MRD-MIS-005	Yes	It's been purchased through UKSEDS
NRC RS-MRD-MIS-006	Yes	The motor form was submitted
NRC RS-MRD-MIS-007	Partially	Unable to determine weight at the moment due to delays in shipment





NRC RS-MRD-MIS-008	Partially	To be tested
NRC RS-MRD-CPD-001	Yes	5V is being used
NRC RS-MRD-CPD-002	Yes	The power of the battery supports the system for approximately 1 hour 30 minutes
NRC RS-MRD-CPD-003	Yes	Current of 1.5 amps will power the Customers Payload.
NRC RS-MRD-CPD-004	Partially	Due to unclear weight, position of the port is undecided.
NRC RS-MRD-CPD-005	Yes	Provided by the launch service
NRC RS-MRD-CPD-006	Partially	Adapters are designed and printed. Once position of port is decided, the right adapter will be chosen.
NRC RS-MRD-CPD-007	Yes	Payload bay has measurements are 45mm diameter and 53mm length.
NRC RS-MRD-CPD-008	Yes	Design of rocket and integrated payload accounts for this
NRC RS-MRD-CPD-009	Partially	Design is still being discussed to figure out how this will be achieved.
NRC RS-MRD-CPD-010	Yes	Theres a hole in the rocket for the camera.
NRC RS-MRD-CPD-011	Partially	Awaiting the arrival of the bolts
NRC RS-MRD-RPD-001	Partially	Testing uncomplete, awaiting parts
NRC RS-MRD-RPD-002	Yes	Using the barometer on a 10dof IMU





NRC RS-MRD-RPD-003	Partially	Parts are delayed and programming still needs to be done
NRC RS-MRD-RPD-004	Partially	Parts are delayed and programming still needs to be done
NRC RS-MRD-RPD-005	Partially	To be printed
NRC RS-MRD-RPD-006	Yes	Multiple sensor included in the rocket. Still deciding shall be used for this part of the requirement.

Figure 23: Table of NRC Requirements

4.2 Budgets

Mass Budget

Subsystem	Component	Estimated mass (g)	Margin	Notes
Propulsion	Cesaroni Pro29 108G68	120	-	Manufacturer value
	Motor mount + hardware	80	10%	3D printed + screws
Structures	Main airframe (balsa tube)	200	15%	Veneer construction
	Spruce fins (x4)	120	15%	Aerofoil shaped
	Bulkheads / couplers	60	10%	plywood / printed
	Nosecone (PLA)	80	10%	3D printed
Recovery	24" parachute	70	10%	Spherachute
	Shock cords (kevlar)	40	10%	>2m
	Ejection charge hardware	20	10%	Charge wells, fittings





Avionics	Mercury altimeter V1	25	-	Manufacturer value
	Rideshare PCB + IMU + SD	100	-	Estimated
	Battery (7.4V 2.25Ah)	140	-	Li-ion pack
	Wiring + switches	20	10%	Multi-core wiring
Payload	Customer Payload	75	Fixed	Competition defined
TOTAL		955g		

Power Budget

Parameter	Value
Battery capacity	16.65 Wh
Customer Payload	5.63 Wh
Rideshare	~0.75 Wh
Total required	~6.4 Wh
Energy margin	>2.5×

Figure 24: Table of Power Budget

5. Manufacturing Processes

5.1 Launch Vehicle

- The main body will be produced using thin balsa wood veneers which will be wrapped around each other and fixed using wood adhesives to produce a fixed tube design with an internal width of 5cm and thickness of 0.5cm
- The fins will be cut from a large plank of spruce using a wood saw. Then they will be sanded down to produce an aerofoil cross-section with a maximum width of 0.5cm and measured using a calliper. It will be connected to the main body by sliding into cut-outs in the main body and secured using wood adhesives.
- The nosecone will be produced using PLA filaments in a 3D printed and secured to the main body using a tight-fitting shoulder, which will hold the nosecone in place.

5.2 Rideshare Payload





- PCB designed in CAD and fabricated externally with SMD assembly using PCBWay.
- Avionics sled manufactured via 3D printing.
- UKSEDS launch vehicle adapter 3D printed without modification to interface faces.
- Wiring strain-relieved and mechanically secured.

5.3 Issues and Solutions

Issue	Solution
Severe delay of parts for Payload	Created a provisional Circuit board with similar parts
Extremely high power requirement for Customers payload	Using a Buck converter instead of the planned Linear Voltage Regulator
Power outage could cause SD card to corrupt	Fram as a separate data storage
Slow preparation of rocket on Launchpad and slow data retrieval after landing due to payloads being difficult to remove.	Putting payloads in an avionics sled
Because of a larger rocket diameter, the FOV of the Camera on the Customers Payload would be severely impeded	A larger hole in the rocket to account for the cameras FOV

Figure 25: Table of Manufacturing Issues and Solutions

6. Testing

6.1 Test Matrix

There are many different test matrix table formats, but the chosen format should show all the tests you are planning to carry out and the corresponding requirements these tests are trying to verify/validate. [See example.](#)

Test	Requirement
45 min runtime (by simulating a continuous load on the Battery)	CPD-002, ESS-001
5 V regulation under 1.5 A load (to be tested using lab supplies)	CPD-001, CPD-003
Altitude measurement validation (lowering sensor from a high place with a string)	RPD-002
SD logging + plotting (to be tested under different loads and refresh rates)	RPD-003





Apogee display retention (by unplugging the power when displaying a value)	RPD-004
Mechanical integration (by testing if it fits the payload inside while fitting in the rocket)	CPD-007
External arming verification (by testing a screw switch on the battery and if al the systems come online with switching it on)	ESS-002
Testing PCB components on a bread board, to make sure our PCB design works.	NRC RS-MRD-RPD-001
The parachute will have a 1kg weight attached to it to ensure that it works efficiently and will be able to support the weight of the rocket.	RRS-001
The mounting bracket will be tested by shaking the mount and observing its ability to negate any damage.	MOR-002, MOR-003
The fins will be tested by applying weights equal to the main body of the rocket from the tip chord edge longitudinally and laterally	SAR-002 & SAR-003
The main rocket body will be tested by suspending the main body from the centre of mass and applying a weight equal to the total weight of the rocket from the leading end of the main body	SAR-004

Figure 26: Test Matrix Table

6.2 Test Results

For Payloads and Avionics: (To be completed following hardware arrival and integration.)
(due to severe delays.)

7. Final Design

7.1 Design Changes





The rocket design has been modified to achieve a stability margin between 1.75 and 2.50 calibres, in accordance with **NRC RS-TSN-SMN-003 | Rev: A.**

The changes implemented are outlined below:

- **Main Body:** The body material has been changed from cardboard to balsa wood veneers to reduce overall structural density.
- **Nose Cone:** The nose cone geometry has been redesigned to incorporate a shoulder that fits securely into the main body. The material has been changed from polystyrene to PLA, and the component will be manufactured using 3D printing.
- **Fins:** Both the dimensions and material of the fins have been revised. The material has been changed from plywood to spruce to maintain the required stability margin following the other structural modifications .

Payload Integration:

- Increased payload bay diameter to 50 mm
- Finalized 50 Hz sampling rate
- Selected 4 A-rated buck converter
- Confirmed shared 7.4 V 2.25 Ah battery architecture
- Confirmed electrical isolation of recovery system
- Implemented external arming switch

7.2 Improvements and Future Work

For Payload:

- Optional FRAM redundancy
- EMI shielding refinement
- Telemetry integration (subject to Ofcom compliance)

Working on getting a launch speed of 20m/s.





8. Acknowledgements

9. References

- [1] Visual Crossing (2026) *Visual Crossing weather data services*. Available at: <https://visualcrossing.com> (Accessed: 16 February 2026).
- [2] Model Rockets (no date) *Mercury V1 Altimeter & Flight Logger Computer*. Available at: <https://modelrockets.co.uk/electronics-and-payloads/altimeters/mercury-v1-altimeter-and-flight-charts/> (Accessed: 16 February 2026).
- [3] Cesaroni Technology Incorporated (no date) *Pro29 108G68-13A motor*. Available at: <https://cesaroni.net/products/p29-2g/108g68-13a/> (Accessed: 16 February 2026).
- [4] ThrustCurve.org (no date) *Cesaroni 108G68-13A thrust curve*. Available at: <https://www.thrustcurve.org/motors/Cesaroni/108G68-13A/> (Accessed: 16 February 2026).
- [5] Raspberry Pi Ltd (no date) *Raspberry Pi Pico 2*. Available at: <https://www.raspberrypi.com/products/raspberry-pi-pico-2/> (Accessed: 16 February 2026).
- [6] Waveshare (no date) *10 DOF IMU Sensor (Version D)*. Available at: [https://www.waveshare.com/wiki/10_DOF_IMU_Sensor_\(D\)](https://www.waveshare.com/wiki/10_DOF_IMU_Sensor_(D)) (Accessed: 16 February 2026).
- [7] Waveshare (no date) *2.9 inch e-Paper module*. Available at: <https://www.waveshare.com/2.9inch-e-paper-module.htm> (Accessed: 16 February 2026).
- [8] Adafruit (no date) *Adafruit micro-SD breakout board tutorial*. Available at: <https://learn.adafruit.com/adafruit-micro-sd-breakout-board-card-tutorial/introduction> (Accessed: 16 February 2026).
- [9] Adafruit (no date) *DHT22 temperature and humidity sensor (Product ID: 385)*. Available at: <https://www.adafruit.com/product/385#technical-details> (Accessed: 16 February 2026).

Appendices

